

Multiple Choice (10 marks)

1. Which situation is an example of an inherited trait?
 - (a) lions preying on zebras
 - (b) monkeys using twigs to get food
 - (c) birds following migratory patterns
 - (d) bears opening coolers at campsites
2. What do the Milky Way Galaxy and other galaxies in the universe have in common?
 - (a) They are similar in shape.
 - (b) They rotate in the same direction.
 - (c) They contain the same number of stars.
 - (d) They have similar elements.
3. Most electrical wires are covered with plastic or rubber. The wires are covered with plastic or rubber because those materials
 - (a) are conductors of electricity
 - (b) make complete electric circuits
 - (c) are not conductors of electricity
 - (d) make the electricity move quickly
4. The nearest star to the Sun is described as being 4.3 light-years away. Which statement explains why the light-year unit of measure is used in this description?
 - (a) The distance is extremely large.
 - (b) The distance is a rough estimate.
 - (c) Only light can travel between stars.
 - (d) Light signals were timed reflecting from the star.
5. As the human population grows, the demand for natural resources increases. Which activity wastes a natural resource?
 - (a) turning off lights
 - (b) using public transportation
 - (c) throwing away aluminum cans
 - (d) opening windows to cool a house

True / False (10 marks)

Answer True or False

6. True or False: Sorting cans by size is the most effective method for separating steel from aluminum cans.
7. True or False: A plastic spoon is not made from a material grown in nature.
8. True or False: Fossil fuels contain chemical energy that is released when they are burned.
9. True or False: The stages of a plant's growth and development follow the order: seed, young plant, and mature plant.
10. True or False: Heat is measured in joules, which is the same unit used for power.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Describe the scientific reasons behind Alex's observation of the sunset from his porch. Discuss how the Earth's rotation and position affect the visibility of the sun.
12. Discuss why the output energy of a machine is always less than the input energy, considering concepts such as energy loss, efficiency, and real-world applications.

End of Paper